



Mass-Storage Structure-II

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Disk Management: Bad Blocks

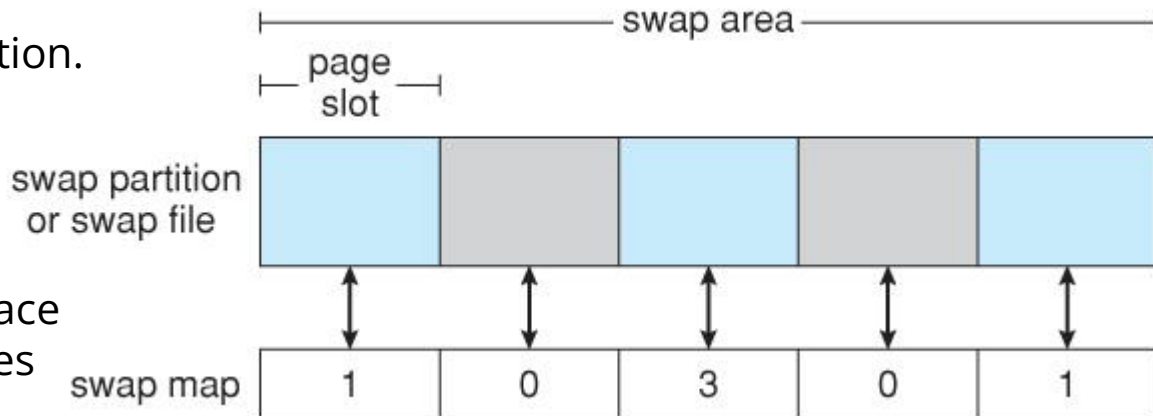
- Disks are prone to failure -> Disks need to be replaced -> Backup
 - Disk can have bad blocks from factory itself
- Disks with IDE controller -> Handled manually
 - Logical formatting by MS-DOS -> scans the disk for bad block -> Special value not to use
 - Data is lost
- Sophisticated disk like SCSI -> Controller maintain a list of bad blocks
 - Initialized at low level formatting and updated so on.
 - **Sector sparing / Forwarding** -> keep spare sectors
 - Controller calculate ECC -> Bad sector -> Report to OS -> Next reboot replace it with spare
 - Few spare sector each cylinder and spare cylinder
- **Sector slipping** -> logical block 17 is bad and spare is at 202
 - Slip them one space -> 201 to 202, 200 to 201 .. so on
- Hard error -> Losing of data

Swap Space Management

- Swapping -> Entire process between disk and memory -> obsolete
 - Modern systems typically swap out pages as needed, rather than swapping out entire processes. Hence the swapping system is part of the virtual memory management system.
- Swap space management -> Low level task of OS -> Best throughput
- **Swap Space Use** -> Vary depending on physical memory, amount of virtual memory and how it is used.
- Safer to overestimate than underestimate -> run out of swap space -> abort processes -> crash entirely -> overestimate waste space
- Set aside some swap space -> Some systems support multiple swap spaces on separate disks in order to speed up the virtual memory system.
- **Swap Space Location** -> normal file system or separate disk or both
- As a large file part of regular filesystem -> Easy to implement, inefficient.
 - Not only must the swap space be accessed through the directory system, the file is also subject to fragmentation issues.
 - Caching block location helps in finding the physical blocks, but that is not a complete fix.

Swap Space Management Contd.

- Swap space as raw partitions -> no file system or directory -> swap space storage manager allocate and deallocate -> optimize speed
- Rather than entire process -> paging
- A process executes -> page containing code are brought from file system -> Thrown away for page out -> Efficient to reread than swap
 - Swap space backing store for Anonymous memory (Solaris) -> Stack, Heap, uninitialized data
- Changes -> allocate swap space when page out rather than creation.
- Linux-> Anonymous or shared
- Swap area -> regular file or raw
- Page slots and Swap Map
- Counter 0 -> available
- Number of mapping to swap space
 - 3-> three different processes

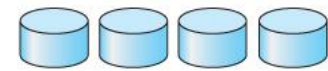


RAIDS

- **Redundant Array of Inexpensive Disks** -> bunch of cheap small disks
 - Independant -> large possibly expensive disks in now a days
- Employ a group of hard drives together with some form of duplication, either to increase reliability or to speed up operations, or both.
- Improvement of Reliability via Redundancy
 - Increasing disks on a system actually *decreases* the Mean Time To Failure, MTTF of the system.
 - Even if disk fails data is not lost -> Duplicate every Disk -> **Mirroring** -> Every write -> two blocks
 - Mean time to data loss = mean time to failure + mean time to repair
 - Does not save from act of God
 - Power failure -> write one copy first then the other -> Nonvolatile RAM Cache
- Improvement in Performance via Parallelism
 - Rate for Read request can be doubled for Mirroring -> Improve transfer rate too
 - **Data stripping** -> splitting the bits of each byte across multiple disks -> **Bit-level Striping**
 - Each access can read 8 times as many data as of single disk. But multiple of 8.
 - **Block-level Striping** -> Blocks of files strip across the disks
 - Parallelism -> Increase Throughput for small access + Reduce response time for large access

RAIDS Levels

- Mirroring high reliability, but expensive
 - Striping high transfer rate but not improve reliability
- RAID Level 0 - Block level stripping, no redundancy
 - Monitoring or parity bit
- Level 1 - includes mirroring only, no striping.
- Level 2 - Memory-Style ErrorCorrectingCode Orgnzn.
 - Single bit error by Parity Bit -> ECC store more bits to reconstruct the data by subtraction from the remaining undamaged data
- Level 3 - Bit-interleaved Parity Organizations
 - Single parity bit for correction and detection -> If the parity of remaining bit = stored parity -> Missing bit 0 otherwise 1
 - Transfer rate is fast but Fewer I/O Per second
- Level 4 - Block-Interleaved Parity Organization
 - similar to level 3, employing block-level striping.
- Level 5 - Block Interleaved Distributed Parity
 - Spreading data and parity among all N+1 disks



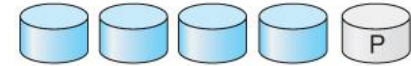
(a) RAID 0: non-redundant striping.



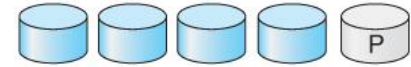
(b) RAID 1: mirrored disks.



(c) RAID 2: memory-style error-correcting codes.



(d) RAID 3: bit-interleaved parity.



(e) RAID 4: block-interleaved parity.



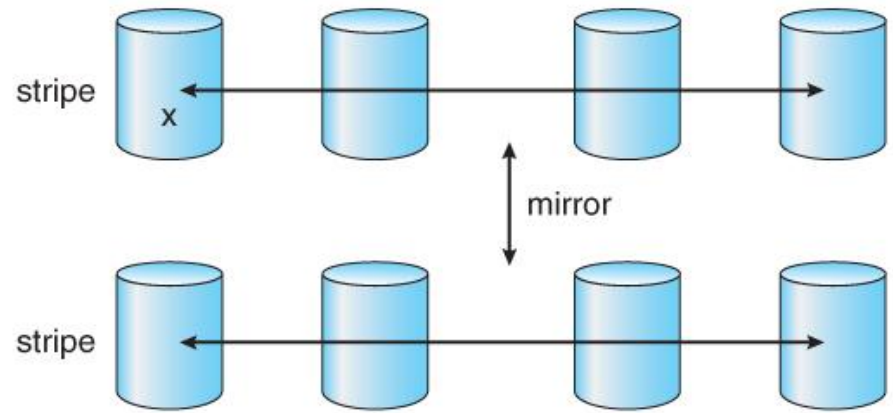
(f) RAID 5: block-interleaved distributed parity.



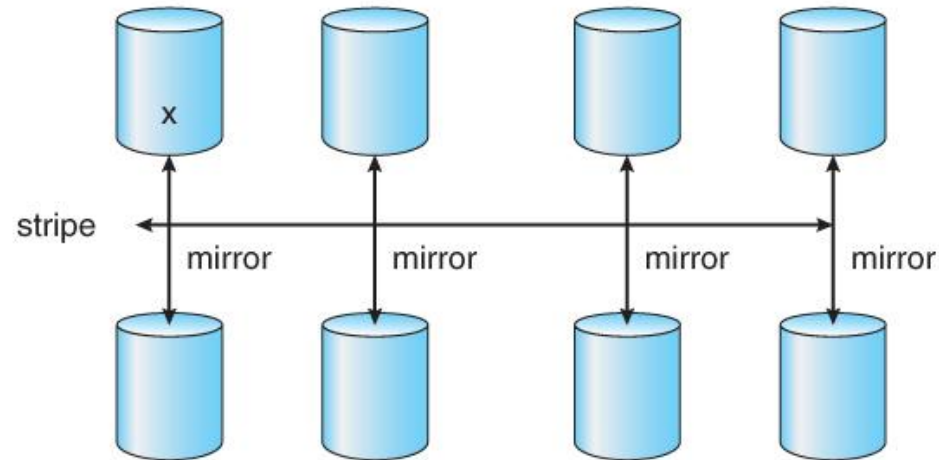
(g) RAID 6: P + Q redundancy.

RAIDS Levels

- Level 6 - P+Q Redundancy Scheme
 - Extra redundant info -> multiple disk failures
- Level 0+1 - Combination of RAID 0 and RAID 1
 - Disks are first striped, and then the striped disks mirrored to another set.
 - 8 disks have been divided into two sets of four
- Level 1+0 - 8 disks are divided into four sets of two
 - mirrors disks in pairs, and then stripes the mirrored pairs.



a) RAID 0 + 1 with a single disk failure.



b) RAID 1 + 0 with a single disk failure.

RAIDS

- Implementation of RAID
 - Within kernel, system software / host bus-adapter / storage array /SAN Interconnect layer
 - Replication -> Automated Duplication of each write. Synchronous -> written locally or remote to make write complete. Asynchronous -> Group together and written periodically
 - Hot Spare Disk -> Configured to be used as a replacement
- Selecting a RAID Level
 - Rebuilding -> Time needed to rebuild in case disk fail -> RAID Level 1
 - Data loss not critical -> high performance application -> RAID Level 0
 - RAID 1+0 or 0+1 -> both performance and reliability is important
 - Storing large data -> RAID Level 5
- Extensions - Array of tapes, Data over wireless system
- Problems with RAID
 - Protect against physical media error, not other hardware or software errors
 - Internal checksums of all blocks -. Higher level of consistency, error detection and error corrections

Stable Storage

- A storage medium in which data is *never* lost, even in the face of equipment failure in the middle of a write operation.
 - To implement this requires two (or more) copies of the data, with separate failure modes.
 - A failure during an update -> not all copies in damaged state
- An attempted disk write results in one of three possible outcomes:
 - The data is successfully and completely written.
 - The data is partially written, but not completely. The last block written may be garbled.
 - No writing takes place at all.
- System needs to detect and invoke recovery procedure -> Consistent state
 - Write the data to the first physical block.
 - After step 1 had completed, then write the data to the second physical block.
 - Declare the operation complete only after both physical writes have completed successfully.
- During recovery each pair of physical blocks are examined
 - If both blocks are identical and there is no sign of damage, then no further action is necessary.
 - If one block contains a detectable error but the other does not, then the damaged block is replaced with the good copy. (This will either undo the operation or complete the operation, depending on which block is damaged and which is undamaged.)
 - If neither block shows damage but the data in the blocks differ, then replace the data in the first block with the data in the second block.
- NVRAM as cache -> as writing to disk is time consuming

Disk Cache

- A disk cache (cache memory) is a temporary holding area in the hard disk or random access memory (RAM) where the computer stores information that used repeatedly.
 - Speed up the process of storing and accessing information
- The term disk cache can also refer to a disk buffer and cache buffer.
- Working with information stored in memory (RAM) is much faster than working with information stored on disk
 - Disk cache is a software utility that works by reserving a section of memory where it keeps a copy of information that previously read from your disks.
- Since related data is often physically adjacent on a disk, the cache may also make a copy of information near the data that was previously used
- Limited Space -> only a fraction of the information on your disk is in the cache at any one time.
- Some disk caching utilities also cache files you want to save, as well as other information your computer is trying to store on the disk.

Tertiary Storage (Optional)

- Build with removable media -> floppy, tapes, CD, DVD , etc.
- Removable Disk -> Removable magnetic disks nearly as fast as hard drives, but are at greater risk for damage due to scratches.
 - *Optical disks* do not use magnetism at all, but instead use special materials that can be altered (by lasers) to have relatively light or dark spots
 - A *magneto-optical* disk uses a magnetic disk covered in a clear plastic coating that protects the surface
 - The original version of WORM (*Write-Once Read-Many*) drives involved a thin layer of aluminum sandwiched between two protective layers of glass or plastic
- Tapes -> Tape drives typically cost more than disk drives, but the cost per MB of the tapes themselves is lower.
 - Used today for backups, and for enormous volumes of data stored by certain scientific establishments.
- Future Technology ->
 - *Solid State Disks, SSDs*, are becoming more and more popular.
 - *Holographic storage* uses laser light to store images in a 3-D structure, and the entire data structure can be transferred in a single flash of laser light.
 - *Micro-Electronic Mechanical Systems, MEMS*, employs the technology used for computer chip fabrication to create VERY tiny little machines.